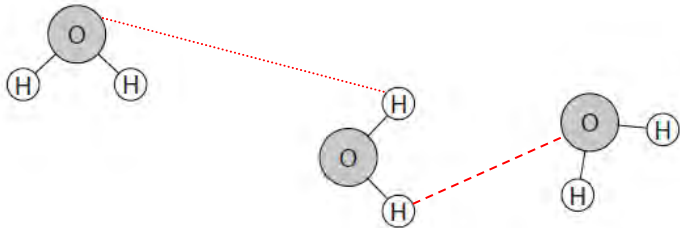
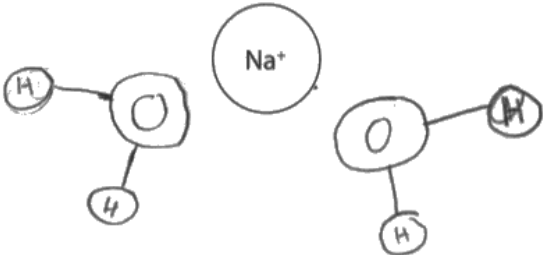


Question number	Answer	Additional guidance	Mark
4(a)(i)	An explanation that includes the following points: - the oxygen (atom) is slightly negative and the hydrogens (atoms) are slightly positive} (1) {there is (an uneven) charge distribution (across the molecule) / uneven distribution of electrons / the oxygen pulls the electrons towards it} (1)	Penalise ref to molecules once ACCEPT δ^- / δ^+ from a diagram ACCEPT comparisons of electronegativity unbalanced charge oxygen has more {protons / more positive nucleus} electrons closer to the oxygen	(2)

Question number	Answer	Additional guidance	Mark
4(a)(ii)	a line drawn between an O of one molecule and a H of another molecule (1)	 If more than one H bond shown then they must be both correct and using different atoms on any one molecule If other water molecules draw then mark if correct	(1)

Question number	Answer	Additional guidance	Mark
4(b)	water molecules clustered around the Na^+ with the O facing the Na^+ and the Hs facing away (1)	NB if charges are shown, they must be correct and partial ACCEPT any number of water molecules but all must be correct DO NOT ACCEPT circles overlapping solid lines joining circles 	(1)

Question number	Answer	Additional guidance	Mark
4(c)	A description that includes four of the following points: • (A) (increase in temperature can) (linear) increase the solubility (1) • (B) (increase in temperature (up to 32°C) can) (exponentially) increase the solubility and then decrease it (1) • (C) (increase in temperature can have) a slight increase on the solubility (1) • (D) (increase in temperature can) decrease the solubility (1) • statement referring to different chemicals have different solubilities at different temperatures (1)	ACCEPT positive correlation ACCEPT no effect ACCEPT negative correlation	(4)